

DS Unit 1 — Arrays, Linked Lists & Complexity

Complete Notes

Data Structures (DS) · Unit 1 · Arrays, Linked Lists & Complexity · 2025

Overview

These notes cover **Arrays, Linked Lists & Complexity**. Core data structures and the algorithms built on them — arrays, lists, stacks, trees, graphs, sorting and hashing. Each section states the core idea, a worked intuition and the points examiners look for.

1. Asymptotic Notation

Big-O, Omega and Theta describe upper, lower and tight bounds on an algorithm's growth rate.

Why it matters: Asymptotic Notation appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of asymptotic notation in one line.
- A small example showing asymptotic notation in practice.
- Common mistakes students make and how to avoid them.

2. Arrays

Arrays store elements contiguously, giving $O(1)$ indexed access but $O(n)$ insertion in the middle.

Why it matters: Arrays appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of arrays in one line.
- A small example showing arrays in practice.
- Common mistakes students make and how to avoid them.

3. Singly Linked Lists

Each node stores data and a pointer to the next node, allowing $O(1)$ insertion at the head.

Why it matters: Singly Linked Lists appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of singly linked lists in one line.
- A small example showing singly linked lists in practice.
- Common mistakes students make and how to avoid them.

4. Doubly and Circular Linked Lists

Doubly linked lists keep prev and next pointers; circular lists link the tail back to the head.

Why it matters: Doubly and Circular Linked Lists appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of doubly and circular linked lists in one line.
- A small example showing doubly and circular linked lists in practice.
- Common mistakes students make and how to avoid them.

5. Sparse Matrices

Sparse matrices store only non-zero entries using triplet or linked representations.

Why it matters: Sparse Matrices appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of sparse matrices in one line.
- A small example showing sparse matrices in practice.
- Common mistakes students make and how to avoid them.

6. Polynomial Representation

Linked lists represent polynomials as sequences of coefficient-exponent terms for addition and multiplication.

Why it matters: Polynomial Representation appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of polynomial representation in one line.
- A small example showing polynomial representation in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Asymptotic Notation · Arrays · Singly Linked Lists · Doubly and Circular Linked Lists · Sparse Matrices · Polynomial Representation